Research Interests and Student Opportunities

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I am broadly interested in collaborating on and/or supervising projects that align with one or more of the following areas:

1. **Large Language Models**
   - Evaluation methodologies
   - Fine-tuning strategies
   - Benchmarking
   - Explainability
   - Hallucinations mitigation
2. **Knowledge Graphs**
   - Construction (rule-based, automated, or hybrid)
   - Graph repair (error detection and correction)
   - Knowledge-graph-based question answering
3. **Information Retrieval**
   - Traditional retrieval methods for language-specific or domain-specific corpora
   - Retrieval augmented generation (RAG), including text-based and graph-based variants
   - Generation-supported retrieval pipelines
   - Multilingual Indian information retrieval systems
4. **Computational Linguistics**
   - Empirical and theoretical exploration of linguistic phenomena in Indian languages, including but not limited to tasks such as
     - Dependency parsing
     - Named entity recognition
     - Grammatical error correction
     - Lexical or semantic grouping
     - Discourse analysis
   - Evaluation of existing, and development of new metrics more suited for various linguistic tasks in Indian languages
5. **Sanskrit Computational Linguistics**
   - Any task centered around Sanskrit, including but not limited to the above-mentioned areas
6. **NLP-Oriented Tool Building**
   - Development of web-based systems/Python libraries to support NLP workflows
   - Annotation tools
   - Agentic systems

7. **Creative Text Generation**
    - Generation of poetry, genre-specific prose, and other forms of creative writing
    - Generating novel content, devising metrics for evaluation of the same

Indian Language NLP
- Any task centered around one or more Indian languages

Note

1. These areas often intersect, and interdisciplinary proposals are welcome.
2. My primary focus is on Sanskrit, followed by other Indian languages (in addition to Hindi), and then English.
3. Projects involving only English are generally discouraged.

Useful Prerequisites

If you are a student,

Having prior exposure to some or many of the following concepts and skills will be useful for getting started faster.

If you have a Computer Science background,

1. Foundations of Deep Learning: An understanding of neural networks, the intuition behind backpropagation, and common evaluation practices
2. Transformers and Core NLP Models: Familiarity with transformer architectures, language modeling objectives, and sequence-to-sequence task formulations
3. Practical Experience with LLMs: Hands-on experience running language models locally (e.g., programmatic inference, loading models, working with tokenizers) using frameworks such as PyTorch, Hugging Face, or similar toolkits (ollama, lang-chain)
4. Familiarity with usage of tools such as LaTeX, Git, formats such as JSON, YAML, experience with Python, Jupyter notebooks

If you have a Sanskrit background,

1. Knowledge of Sanskrit enough to hold a small conversation, read simple texts, and understand basic grammar
2. Traditional training in Sanskrit grammar (vyākaraṇa) or other śastra such as chanda, gaṇita, nyāaya
3. Familiarity with computational approaches to Sanskrit, including some exposure to Sanskrit processing tools
4. Familiarity with transliteration schemes such as IAST, ISO, Harvard-Kyoto, ITRANS, SLP1, WX